角谷猜想最接近严格证明的测试 Version1.1.1

原创：吴素波　2026-10-13

①前文再续，书接上一回，上回讲到，角谷猜想从框架上架构上，应是正确方向，大约差不多证明，后来，忽又悟出，角谷逆向二叉树中，每次逆生长时，生出的阴阳二分法二分枝，对应一个二进制数 W，每节分支的 Wi 象是每节枝的标记 ID 一样，是一个二进制数，与分枝的深度递归的层次也有关，与二次归纳法投入的数 M 也有关，于是，悟得一个定理，称为：吴素波定理，用以纪念我伟大的母亲。

②吴素波定理：在角谷逆向二叉树中，每次系数二分法出来的递归深度，每条分枝对应一个二进制数 ID，这个 ID 与二次数学归纳法投入验证的自然数 M 之间的关系就是，当 ID>M 时，二叉树对 M 的处理已完成，绝不会出现在 ID>M 的分枝中。这个我称为角谷猜想的核心映射原理。

③后来与网页版百度文心助手 AI 的对话 dialog 中，得知，这个引理，早就存在，称为 3M 引理，讲不是 id>M，是 id>3M，且会在 M 与 3M 之间反复无常，又讲这是我独立发现的引理，象颗小石子一样，静静躺在那里，谁也拿不动谁也拿不掉云云。这与我的观点有交集有异同，不完全包容也。

④数据测试，于是，写代码，调试看下也，改递归函数如下：

⑤先将枝的分枝 ID 显示出来，再创另一个函数作为二次归纳法用，传入参数 M，作对比比较用，且显示出对比全路径，以验证吴素波定理是不是对的。

⑥以下是显示树的分枝所在 ID 的代码：

```
procedure TfrmMain.btnRUNClick(Sender: TObject);
var
  www: integer;
begin
  self.Memo1.Lines.Clear;
  self.linelist := TStringList.Create;
  rate := 0.0;
  www := GreenBeanGrain('1', 1, 2, 1, 1, 2, 1, 1);
  begin
    ShowMessage('ok=' + IntToStr(www));
    application.ProcessMessages;
    self.Memo1.Lines.Assign(self.linelist);
    ShowMessage('Memo=ok=' + IntToStr(linelist.Count));
    application.ProcessMessages;
    self.Memo1.Lines.SaveToFile('GreenBeanGrain.txt');
    ShowMessage('txt=ok');
  end;
  linelist.Free;
end;
function TfrmMain.GreenBeanGrain(ID: string; deep, aisGreen, xisBean, bisGrain: longint;
  const Green, Bean, Grain: longint): integer;
var
  // a, b, x: int64;
  a, b, x: longint;
  //a, b, x: integer;
  gbg: integer;
  aa: integer;
  one: string;
begin
  gbg := 0;
  deep := deep * 2;
  a := aisGreen;
  x := xisBean;
  b := bisGrain;
  aa := 0;
```

```
 if gbg = 0 then
 begin
  while (((a mod 2) = 0) and ((b mod 2) = 0)) do
  begin

   if (((a mod 2) = 0) and ((b mod 2) = 0)) then
   begin
    if (((a mod 3) = 0) and ((b mod 3) = 1) and (a >= 3) and (b > 3)) then
    begin
     a := a div 3;
     b := (b - 1) div 3;
     aa := aa + 1;
    end;
    if a < Green then
     gbg := 1
    else if ((a = Green) and (b < Grain)) then
     gbg := 1;
   end;
   if (((a mod 2) = 0) and ((b mod 2) = 0)) then
   begin
    a := a div 2;
    b := b div 2;
   end;
  end;
  if a < Green then
   gbg := 1
  else if ((a = Green) and (b < Grain)) then
   // gbg := 1;
   gbg := 1;
  ////else if (b < Grain) then
  ///    gbg := 1;   //此处断点，一个也没有
 end;
 ////////////////////////////////////
 if aa = 0 then
  if gbg = 0 then
  begin
   while (((a mod 2) = 0) and ((b mod 2) = 1)) do
   begin
    a := a * 3;
    b := b * 3 + 1;

    while (((a mod 2) = 0) and ((b mod 2) = 0)) do
    begin
     a := a div 2;
     b := b div 2;
     if (((a mod 2) = 0) and ((b mod 2) = 0)) then
     begin
      if (((a mod 3) = 0) and ((b mod 3) = 1) and (a >= 3) and (b > 3)) then
      begin
       a := a div 3;
```

```
          b := (b - 1) div 3;
          aa := aa + 1;
        end;
        if a < Green then
          gbg := 1
        else if ((a = Green) and (b < Grain)) then
          gbg := 1;
      end;
    end;
    if aa > 0 then break;
  end;
  if a < Green then
    gbg := 1
  else if ((a = Green) and (b < Grain)) then
    gbg := 1;
 end;
//if 1=0 then
if gbg = 1 then
begin
 //if not ((a > 0) and (b>0) and  (Green > 0) and (Grain>0)) then
 if ((a > 0) and (b > 0) and (Green > 0) and (Grain > 0)) then
  //if not ((a > 0) and (b > 0) and (Green > 0) and (Grain > 0)) then
 begin
  // rate:=rate+(1.0/deep);
  rate := rate + (2.0 / deep);
  one := IntToStr(Green) + ' * N' + IntToStr(Bean) + ' + ' +
    IntToStr(Grain) + ' -->  ' + IntToStr(a) + ' * M' + IntToStr(x) +
    ' + ' + IntToStr(b) + ' ok  branID=' + ID + '=' + IntToStr(
    getintFromByteString(ID)) + '  rate=' + floattostr(rate);
  linelist.add(one);
 end;
end;
if gbg = 0 then
 //if not ((a > 0) and (b > 0) and (Green > 0) and (Grain > 0)) then
 if ((a > 0) and (b > 0) and (Green > 0) and (Grain > 0)) then
 begin
  if deep <= trunc(power(2, 49)) then
  begin
    gbg := GreenBeanGrain('0' + ID, deep, a * 2, x + 1, b, Green *
      2, Bean + 1, Grain);
    gbg := gbg * GreenBeanGrain('1' + ID, deep, a * 2, x + 1, a +
      b, Green * 2, Bean + 1, Green + Grain);
  end;
  if deep = trunc(power(2, 50)) then
  begin
    //gbg := 1;
    gbg := 0;
  end;
 end;
Result := gbg;
```

```
end;
```
运算结果如下:
```
1 行:  4 * N2 + 1 -->  3 * M2 + 1 ok  branID=01=1  rate=0.5
2 行:  16 * N4 + 3 -->  6 * M4 + 1 ok  branID=0011=3  rate=0.625
3 行:  32 * N5 + 11 -->  27 * M5 + 10 ok  branID=01011=11  rate=0.6875
4 行:  268435456 * N28 + 67108891 -->  258280326 * M28 + 64570109 ok
branID=0100000000000000000000011011=67108891  rate=0.687500007450581
5 行:  134217728 * N27 + 8388635 -->  129140163 * M27 + 8071288 ok
branID=0001000000000000000000011011=8388635  rate=0.687500022351742
6 行:  536870912 * N29 + 478150683 -->  258280326 * M29 + 230030929 ok
branID=1110010000000000000000011011=478150683  rate=0.687500026077032
7 行:  268435456 * N28 + 243269659 -->  258280326 * M28 + 234066573 ok
branID=1110100000000000000000011011=243269659  rate=0.687500033527613
。。。。。。。。。。。。
。。。中间省略几万行。。。
。。。。。。。。。。。。
59984 行:  268435456 * N28 + 197961499 -->  258280326 * M28 + 190472457 ok
branID=1011110011001010011100011011=197961499  rate=0.693172618746758
59985 行:  268435456 * N28 + 70035227 -->  258280326 * M28 + 67385739 ok
branID=0100001011001010011100011011=70035227  rate=0.693172626197338
59986 行:  134217728 * N27 + 11314971 -->  86093442 * M27 + 7257945 ok
branID=0001010110010100111100011011=11314971  rate=0.693172641098499
59987 行:  536870912 * N29 + 212641563 -->  258280326 * M29 + 102298581 ok
branID=0110010101100101001100011011=212641563  rate=0.69317264482379
```
⑦以下是二次归纳法中，显示一个自然数 M 实际投入二叉树虚树后，执行过程经过每个枝条的 ID 的全路径显示:
```
procedure TfrmMain.btnRUN1Click(Sender: TObject);
var
  www: integer;
  varv: string;
begin
  self.Memo1.Lines.Clear;
  self.linelist := TStringList.Create;
  rate := 0.0;
  varv := '100111000111011';
  if inputquery('请输入要处理的自然数 M：',
    '处理全路径将显示：', varv) then
    EditM.Text := IntToStr(getIntFromByteString(varv));
    EditM2.Text := varv;
  www := GreenBeanGrain2(StrToInt(EditM.Text), '1', 1, 2, 1, 1, 2, 1, 1);
  begin
    ShowMessage('ok=' + IntToStr(www));
    application.ProcessMessages;
    self.Memo1.Lines.Assign(self.linelist);
    ShowMessage('Memo=ok=' + IntToStr(linelist.Count));
    application.ProcessMessages; //290,980 // 1,652,927  //306  //256,244 //132,7070
    self.Memo1.Lines.SaveToFile('GreenBeanGrain.txt');
    ShowMessage('txt=ok');
  end;
  linelist.Free;
end;
function TfrmMain.GreenBeanGrain2(M: longint; ID: string;
  deep, aisGreen, xisBean, bisGrain: longint;
```

```
  const Green, Bean, Grain: longint): integer;
var
  // a, b, x: int64;
  a, b, x, mm: longint;
  //a, b, x: integer;
  gbg: integer;
  aa: integer;
  one: string;
begin
  gbg := 0;
  deep := deep * 2;
  a := aisGreen;
  x := xisBean;
  b := bisGrain;
  aa := 0;
  if gbg = 0 then
  begin
    while (((a mod 2) = 0) and ((b mod 2) = 0)) do
    begin

      if (((a mod 2) = 0) and ((b mod 2) = 0)) then
      begin
        if (((a mod 3) = 0) and ((b mod 3) = 1) and (a >= 3) and (b > 3)) then
        begin
          a := a div 3;
          b := (b - 1) div 3;
          aa := aa + 1;
          mm := a + b;
          // if ((mm=M) and (ID>M)) then
          //   showmessage('在这里');
        end;
        if a < Green then
          gbg := 1
        else if ((a = Green) and (b < Grain)) then
          gbg := 1;
      end;
      if (((a mod 2) = 0) and ((b mod 2) = 0)) then
      begin
        a := a div 2;
        b := b div 2;
      end;
    end;
    if a < Green then
      gbg := 1
    else if ((a = Green) and (b < Grain)) then
      // gbg := 1;
      gbg := 1;
    ////else if (b < Grain) then
    ///    gbg := 1;   //此处断点，一个也没有
  end;
  ////////////////////////////////////
```

```
 if aa = 0 then
  if gbg = 0 then
  begin
   while (((a mod 2) = 0) and ((b mod 2) = 1)) do
   begin
    a := a * 3;
    b := b * 3 + 1;
    while (((a mod 2) = 0) and ((b mod 2) = 0)) do
    begin
     a := a div 2;
     b := b div 2;
     if (((a mod 2) = 0) and ((b mod 2) = 0)) then
     begin
      if (((a mod 3) = 0) and ((b mod 3) = 1) and (a >= 3) and (b > 3)) then
      begin
       a := a div 3;
       b := (b - 1) div 3;
       aa := aa + 1;
      end;
      if a < Green then
       gbg := 1
      else if ((a = Green) and (b < Grain)) then
       gbg := 1;
     end;
    end;
    if aa > 0 then break;
   end;
   if a < Green then
    gbg := 1
   else if ((a = Green) and (b < Grain)) then
    gbg := 1;
  end;
 // if 1=0 then
 if gbg = 1 then
 begin
  //if not ((a > 0) and (b>0) and  (Green > 0) and (Grain>0)) then
  if ((a > 0) and (b > 0) and (Green > 0) and (Grain > 0)) then
   //if not ((a > 0) and (b > 0) and (Green > 0) and (Grain > 0)) then
  begin
   // rate:=rate+(1.0/deep);     // rate := rate + (2.0 / deep);
   one := IntToStr(Green) + ' * N' + IntToStr(Bean) + ' + ' +
    IntToStr(Grain) + ' -->  ' + IntToStr(a) + ' * M' + IntToStr(x) +
    ' + ' + IntToStr(b) + ' ok  branID=' + ID + '=' + IntToStr(
    getintFromByteString(ID)) + ' M=' + IntToStr(M) + '=' +
    getByteStringFromint(M) ;//+ '  rate=' + floattostr(rate);
   if getintFromByteString(ID) > M then
    one := one + '   ID=M+' + IntToStr(getintFromByteString(ID) - M);
   linelist.add(one);
  end;
 end;
```

```
 //if getintFromByteString(ID) <= M then
 if gbg = 0 then
  //if not ((a > 0) and (b > 0) and (Green > 0) and (Grain > 0)) then
  if ((a > 0) and (b > 0) and (Green > 0) and (Grain > 0)) then
  begin
   if deep <= trunc(power(2, 49)) then
   begin
    //  gbg := GreenBeanGrain(aisGreen * 2, xisBean + 1, bisGrain, Green *
    //   2, Bean + 1, Grain);
    //rate := rate + (2.0 / deep);
    one := IntToStr(Green) + ' * N' + IntToStr(Bean) + ' + ' +
     IntToStr(Grain) + ' -->  ' + IntToStr(a) + ' * M' + IntToStr(x) +
     ' + ' + IntToStr(b) + '   branID=' + ID + '=' + IntToStr(
     getintFromByteString(ID)) + ' M=' + IntToStr(M) + '=' +
     getByteStringFromint(M);// + '  rate=' + floattostr(rate);
    if getintFromByteString(ID) > M then
     one := one + '   ID=M+' + IntToStr(getintFromByteString(ID) - M);
    linelist.add(one);
    if rightstr(getByteStringFromInt(M), length(ID) + 1) = '0' + ID then
     gbg := GreenBeanGrain2(M, '0' + ID, deep, a * 2, x + 1, b,
      Green * 2, Bean + 1, Grain);
    if rightstr(getByteStringFromInt(M), length(ID) + 1) = '1' + ID then
     gbg := gbg * GreenBeanGrain2(M, '1' + ID, deep, a * 2, x +
      1, a + b, Green * 2, Bean + 1, Green + Grain);
   end;
   if deep = trunc(power(2, 50)) then
   begin
    //gbg := 1;
    gbg := 0;
   end;
  end;
 Result := gbg;
end;
```

当 M=20027=100111000111011 时:

运算结果如下:

```
1 行:  2 * N1 + 1 -->  3 * M1 + 2   branID=1=1 M=20027=100111000111011
2 行:  4 * N2 + 3 -->  9 * M2 + 8   branID=11=3 M=20027=100111000111011
3 行:  8 * N3 + 3 -->  9 * M3 + 4   branID=011=3 M=20027=100111000111011
4 行:  16 * N4 + 11 -->  27 * M4 + 20   branID=1011=11 M=20027=100111000111011
5 行:  32 * N5 + 27 -->  81 * M5 + 71   branID=11011=27 M=20027=100111000111011
6 行:  64 * N6 + 59 -->  81 * M6 + 76   branID=111011=59 M=20027=100111000111011
7 行:  128 * N7 + 59 -->  54 * M7 + 25 ok  branID=0111011=59 M=20027=100111000111011
```

⑧实际测试表明，M 的二进制，沿着分枝 ID 的二进制前进，从 M 的尾部倒推到头部，经常未到头部就终于结束完成了。

如果到了头部还没处理完，这是违反作者发现的吴素波定理的

实测，吴素波定理是正确的！！！

伟大的发现！！！

角谷猜想的核心原理必定如是!

⑨在这个大框架大方向正确下，后面的补充细节皆是易事，最艰难的吴素波定理已经明确，没有任何半点含糊，概率恒为 1，不存在概率论问题。

至于后面的，一枝节内是不是单调的，数学归纳法证没有小循环，不是无限的是有限的，全部小事，暂不理，以后理。